



Vel Tech
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(Estd. u/s 3 of UGC Act, 1956)
Avadi, Chennai

SCHOOL OF COMPUTING
DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE
Academic year 2025-2026
(Winter Semester)

Course code: 10214AD701

Course Name: Major Project

TTS NO	Supervisor Name	VTU NO.	Student Name1	VTU NO.	Student Name 2
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Title of the Project: SMARTFLOW: Intelligent Traffic Management with Real-Time Density and Emergency Response

Problem Statement

Rapid urbanization has led to severe traffic congestion, increased travel delays, and challenges in ensuring timely emergency response. Conventional traffic signal control systems typically operate using fixed-time cycles and are unable to adapt to dynamic variations in traffic density. This results in prolonged waiting times, excessive fuel consumption, and elevated air pollution levels. Furthermore, emergency vehicles such as ambulances and fire engines often experience significant delays at intersections, potentially endangering lives and reducing overall transportation efficiency. Hence, there is a critical need for an intelligent, real-time traffic management system capable of dynamically adjusting signal timings, alleviating congestion, and providing priority passage for emergency services.

Mapping of the Project to Complex Engineering Problem (CEP)

Level	Attribute	Characteristics	Targeted WKs (wherever applicable)	Justification (as applicable)
WP1	Depth of Knowledge	Cannot be resolved without in-depth engineering knowledge at the level of K3, K4, K5, K6 or K8	WK3, WK4, WK5, WK6, WK8	The secure data management and visualization platform requires deep understanding of engineering fundamentals (distributed databases, networking), specialist knowledge (homomorphic

				encryption, blockchain, IDS), engineering design (multi-layered architecture), professional practice (security enforcement in data centers), and extensive research literature to address scalability and cybersecurity challenges.
WP2	Range of conflicting requirements	Involve wide-ranging and/or conflicting technical and non-technical issues	WK4, WK5, WK6, WK8	The work addresses conflicting requirements such as data security versus real-time performance, privacy versus accessibility, scalability versus computational overhead, and compliance versus usability. Ethical, legal, and societal concerns related to sensitive data protection and future cybersecurity threats are also considered.
WP3	Depth of analysis required	Have no obvious solution and require originality in analysis	WK3, WK4, WK5, WK8	There is no standard solution for secure, scalable, real-time data visualization under zero-trust constraints. The solution requires abstract thinking, original architectural modeling, algorithm selection, and extensive performance evaluation using AI-based threat detection and encrypted computation.
WP4	Familiarity of issues	Involve infrequently encountered or novel problems	WK4, WK8	The integration of homomorphic encryption, blockchain-based audit logging, and AI-driven visualization in data centers represents emerging and infrequently encountered engineering problems beyond routine data management practices.
WP5	Extent of applicable codes	Address problems not encompassed by	WK6, WK8	Existing data center security and visualization standards do not fully address

		standards and codes of practice		encrypted analytics, blockchain-enabled transparency, and zero-trust visualization architectures. Hence, research-based engineering judgment was essential.
WP6	Stakeholder involvement and conflicting requirements	Involve collaboration across disciplines and diverse stakeholders	WK5, WK6, WK7	The system involves administrators, end users, organizations, and regulatory stakeholders with varying requirements. RBAC and blockchain audit logging are used to resolve conflicts related to access control, transparency, and compliance.
WP7	Interdependence	Address high-level problems requiring a systems approach	WK3, WK5, WK6	The platform consists of interdependent subsystems including frontend interfaces, backend processing, distributed databases, AI analytics, encryption modules, and visualization layers, requiring a holistic systems engineering approach.

Mapping of the Project to Complex Engineering Activities (CEA)

EA No.	Attribute	Characteristics	Justification based on the Project
EA1	Range of Resources	Diverse resources and technologies	The project utilizes large-scale datasets, distributed infrastructure, blockchain platforms, AI frameworks, cloud resources, and skilled human roles such as admins, users, and auditors.
EA2	Level of Interactions	Optimization of complex interactions	It resolves interactions between security layers, data processing pipelines, visualization performance, and user access control under heavy workloads.
EA3	Innovation	Creative and research-based solutions	Innovative use of homomorphic encryption, blockchain-based audit logging, AI-powered anomaly detection, and zero-trust architecture demonstrates research-driven engineering practice.
EA4	Consequences to Society	Significant societal impact	Ensures protection of sensitive data in domains like healthcare and finance, reduces cyber risks, and enhances trust in digital infrastructure,

EA No.	Attribute	Characteristics	Justification based on the Project
	and Environment		where failures could have serious societal consequences.
EA5	Familiarity	Extends beyond prior experience	The project applies principle-based approaches and emerging technologies beyond conventional data management systems.

Sustainable Development Goals (SDG) Mapping

SDG No.	SDG Title	Relevance to the Project
SDG 9	Industry, Innovation and Infrastructure	Promotes secure, scalable, and innovative digital infrastructure for next-generation data centers.
SDG 16	Peace, Justice and Strong Institutions	Enhances transparency, data integrity, auditability, and trust through blockchain and strong security mechanisms.
SDG 11	Sustainable Cities and Communities	Supports reliable data infrastructure for smart city and urban analytics applications.
SDG 3	Good Health and Well-being	Enables secure handling and visualization of healthcare data, reducing risks of data breaches and fraud.

Overall Justification: The project clearly satisfies all characteristics of Complex Engineering Problems and Activities by requiring advanced multidisciplinary engineering knowledge, addressing conflicting stakeholder requirements, applying innovative and research-based solutions, and producing significant societal impact. It aligns strongly with national accreditation and OBE/NBA expectations for CEP and CEA classification.

Depth of Knowledge – Required Knowledge Levels (K3–K8)

The problem addressed in this project cannot be resolved without in-depth engineering knowledge spanning the following **Knowledge Profile levels**:

K3 – Engineering Fundamentals

- Understanding of computer networks, operating systems, database fundamentals, and software architecture.
- Application of encryption principles, access control mechanisms, and data flow models in distributed systems.

K4 – Specialist Knowledge

- Advanced knowledge in cybersecurity techniques such as homomorphic encryption, intrusion detection systems (IDS), and zero-trust security models.
- Expertise in distributed databases, blockchain technology, and cloud-based system design.

K5 – Engineering Design

- Design of a multi-layered, microservices-based system architecture integrating security, data processing, and visualization layers.
- Development of role-based access control (RBAC) frameworks and secure data pipelines.

K6 – Engineering Practice

- Practical implementation using modern tools and frameworks such as Django, Node.js, React, PostgreSQL, MongoDB, and Hyperledger.
- Performance optimization, scalability testing, load balancing, caching strategies, and deployment in multi-node environments.

K7 – Engineering Comprehension

- Ability to analyze system behavior under cyberattacks, high user load, and large data volumes.
- Interpretation of system performance metrics, threat detection outputs, and visualization accuracy for decision-making.

K8 – Research Literature and Emerging Technologies

- Use of contemporary research in secure data analytics, privacy-preserving computation, blockchain auditing, and AI-driven threat detection.
- Evaluation and adaptation of recent IEEE research findings to design innovative and non-standard solutions.

Signature

(Dr. R. Anandh)